

# Every case jamovi can show

Lab test, 28 Sep. This sheet uses the button names on the jamovi screen.  $\alpha = .05$ . A sentence in a box is a definition. Copy numbers from the row you decided to use. If jamovi prints  $< .001$ , write  $p < .001$ .

## Words

**Significant** means the p-value is below .05. You reject the null hypothesis. The test found a difference, or a relationship.

**Not significant** means the p-value is above .05. This includes .051, .058, and .150. You fail to reject the null hypothesis. You do not write “there is no difference” or “there is no relationship.” You still report the size.

**F is significant** means the ANOVA’s p-value is below .05. The group means, or the condition means, are not all equal. F does not name which pair differs.

**F is not significant** means the ANOVA’s p-value is above .05. Stop. Do not interpret a pair as the winner.

**Levene’s p above .05** means the homogeneity test did not find unequal variances. Use the equal-variance row.

**Levene’s p below .05** means jamovi’s homogeneity test suggests the groups have unequal variances. Use the unequal-variance row.

**Q-Q plot met** means the points follow the diagonal line. **Q-Q plot not met** means the points bend, or the ends peel off the line. A large deviation from the diagonal is what jamovi treats as non-normal. One or two end points in a small sample are not enough.

**Normality test** is Shapiro–Wilk. A p below .05 means that test flags the residuals as non-normal. It flags small bends. Write what the Q-Q plot shows.

**Cohen’s d** is the mean difference in SD units. 0.2 small, 0.5 medium, 0.8 large. A value between two of those is “between small and medium” or “between medium and large.” Below 0.2 is smaller than small. Use the absolute value for the size word. The two means give the direction.

**|r|** is the correlation with the minus sign removed.  $\geq .10$  small,  $\geq .30$  medium,  $\geq .50$  large. Below .10 is below the small benchmark. The sign is a separate word: positive or negative.

**$\eta^2$**  is a proportion of variance. 0.71 means 71%. There is no small / medium / large cutoff on the jamovi  $\eta^2$  checkbox.

**A 95% CI for a mean difference includes 0** means that interval agrees with a non-significant t. **Excludes 0** means it agrees with a significant t. The sign must match which mean is higher.

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## Which jamovi screen

| What you can see in the data                                                            | Open this                                                                        |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| You need the mean, the spread, or the shape of one column                               | Exploration → Descriptives                                                       |
| A grouping column with exactly 2 levels, different rows in each level, one score column | T-Tests → Independent Samples T-Test                                             |
| The same people in two score columns                                                    | T-Tests → Paired Samples T-Test                                                  |
| Two score columns, and the question is whether they move together                       | Regression → Correlation Matrix                                                  |
| Same question, plus a third score column to hold constant                               | Correlation Matrix first, then Regression → Partial Correlation                  |
| A grouping column with 3 or more levels, different rows in each level                   | ANOVA → One-Way ANOVA, and also ANOVA → ANOVA for $\eta^2$ and for No correction |
| The same people in 3 or more score columns                                              | ANOVA → Repeated Measures ANOVA                                                  |

Before every analysis, double-click the column header. A grouping column is Nominal. A score column is Continuous. Do not put the ID column into an analysis box.

The outcome score goes in Dependent Variable or Dependent Variables. The group label goes in Grouping Variable or Fixed Factors. A second score, when you are only drawing a residual plot, goes in Covariates.

# Descriptives

**Click.**

Analyses → Exploration → Descriptives. Scores → Variables. If you need a mean for each group, put the group label in Split by. Tick N, Mean, Median, SD, Quartiles. Plots: Histogram, Box plot, Q-Q.

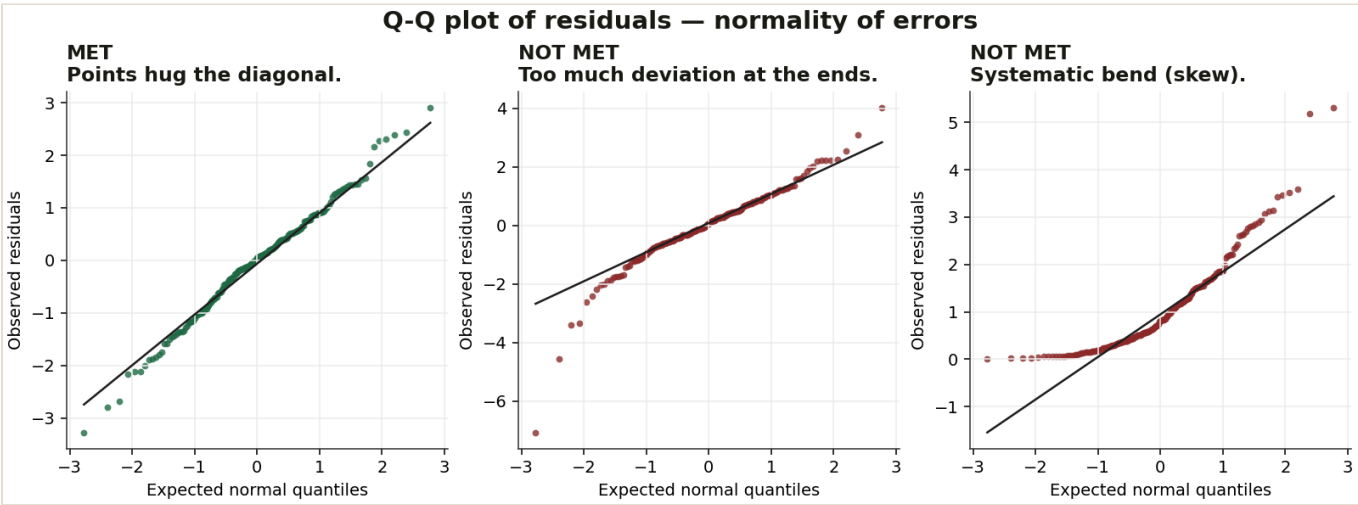
| What you see                   | What it means                                                                                                                                                      | What you write                                                          |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Mean higher than the median    | The right tail is pulling the mean up. Right skew.                                                                                                                 | Name the mean, the SD, and the skew.                                    |
| Mean lower than the median     | Left skew.                                                                                                                                                         | Same, and say left skew.                                                |
| Box from Q1 to Q3, line inside | The box is the middle 50%. The line is the median. IQR = $Q3 - Q1$ .                                                                                               | Quote Q1, median, Q3 if the question is about the box.                  |
| A point beyond the fence       | The fence is $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$ . The whisker stops at the last real score inside the fence. The point beyond the fence is an outlier. | Say the fence, the whisker, and the outlier as three different numbers. |
| Histogram with one bell        | One mode, roughly normal. jamovi also calls this Gaussian.                                                                                                         | Say the shape.                                                          |

# Independent-samples t-test

**Use this screen when** the grouping column has exactly 2 levels and the rows are different people.

**Click.** Analyses → T-Tests → Independent Samples T-Test. Dependent Variables = the score.  
Grouping Variable = the 2-level column. Tick Student's and Welch's. Hypothesis: Group 1 ≠ Group 2, unless the question states a direction. Additional Statistics: Mean difference, Confidence interval, Effect size, Descriptives. Assumption Checks: Homogeneity test, Q-Q plot. The Normality test is optional. The Q-Q plot is the one you describe.

| What you see                                         | What it means                                                                    | Which numbers you copy                                               |
|------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Q-Q points follow the line                           | Normality of residuals is met.                                                   | Write that the Q-Q plot was met.                                     |
| Q-Q points bend or peel off                          | Normality is not met. Say so. The t rows are still on the screen.                | Write that normality was not met, then continue with Levene.         |
| Levene $p > .05$                                     | Equal variances. The equal-variance test is Student's.                           | Copy t, df, p, CI, and d from the Student's row only.                |
| Levene $p < .05$                                     | Unequal variances. The unequal-variance test is Welch's. Its df is not $N - 2$ . | Copy t, df, p, CI, and d from the Welch's row only.                  |
| The chosen row has $p < .05$ , and the CI excludes 0 | The two means differ. Significant.                                               | Name the group with the higher mean.                                 |
| The chosen row has $p > .05$ , or the CI includes 0  | Not significant. Fail to reject the null.                                        | Still write d and its size word. Do not say the groups are the same. |



Q-Q plot. Left: met, points follow the diagonal. Middle: not met, the ends peel off. Right: not met, the points bend (skew). Paired t, one-way ANOVA, and repeated-measures ANOVA use this same picture.

The mean [score] in [group 1] was \_\_ (SD = \_\_), whereas the mean in [group 2] was \_\_ (SD = \_\_). The Q-Q plot of residuals followed the line / bent away from the line, so normality was met / not met. Levene's test was  $F(\\_, \\_) = \\_, p = \\_$ . Because this p is above / below .05, variances are equal / unequal, and Student's / Welch's t was used. The difference was significant / not significant,  $t(\\_) = \\_, p = \\_, 95\% \text{ CI } [\\_, \\_]$ , Cohen's d = \_\_ (small / medium / large / between ... / smaller than small), such that [group] had the higher mean.

Pattern from class: Levene  $p = .125$ , which is above .05, so the Student's row.  $t(31) = 2.12, p = .043$ , which is below .05, so significant.  $d = .74$ , which is between 0.5 and 0.8, so between medium and large.

# Paired-samples t-test

**Use this screen when** two score columns are the same people. This screen has no Grouping Variable, no Homogeneity test, and no Welch’s box.

**Click.** Analyses → T-Tests → Paired Samples T-Test. Put the two columns in one pair. Student’s. Hypothesis: Measure 1 ≠ Measure 2. Additional Statistics: Mean difference, Confidence interval, Effect size, Descriptives. Assumption Checks: Q-Q plot.

| What you see                  | What it means                                                 | What you copy                                             |
|-------------------------------|---------------------------------------------------------------|-----------------------------------------------------------|
| Q-Q follows the line          | Normality of the difference scores is met.                    | Write that.                                               |
| Q-Q bends                     | Normality is not met.                                         | Write that, then still read the t row.                    |
| p < .05 and the CI excludes 0 | The two times differ. Significant. df = number of people – 1. | The higher mean is the direction. d uses 0.2 / 0.5 / 0.8. |
| p > .05 or the CI includes 0  | Not significant.                                              | Still write d.                                            |

The mean of [time 1] was \_\_ (SD = \_\_), whereas the mean of [time 2] was \_\_ (SD = \_\_). The Q-Q plot followed the line / bent, so normality was met / not met. A paired-samples t-test was significant / not significant, t(\_\_) = \_\_, p = \_\_, 95% CI [\_\_, \_\_], Cohen’s d = \_\_ (size), such that the higher mean was at [time].

## Correlation Matrix

**Use this screen when** both columns are continuous scores. It does not compare group means.

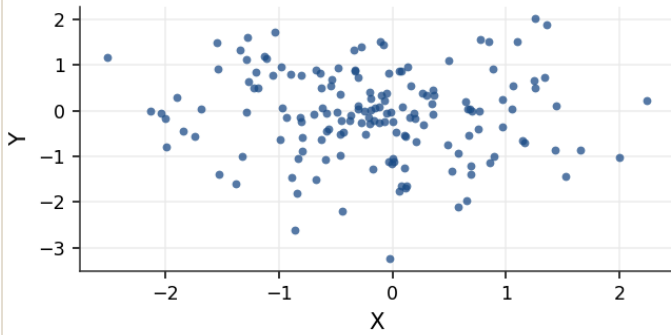
**Click.** Analyses → Regression → Correlation Matrix. Put both columns in. Tick Pearson. Tick Spearman or Kendall's tau-b only after the pictures say Pearson's assumptions failed. Hypothesis: Correlated, unless the question states a direction. Tick Report significance and N.

**Pictures.** Analyses → Regression → Linear Regression. Dependent Variable = Y. Covariates = X. Assumption Checks: Q-Q plot of residuals, Residual plots. Read the pictures. Do not write the regression coefficients unless the question asks for them.

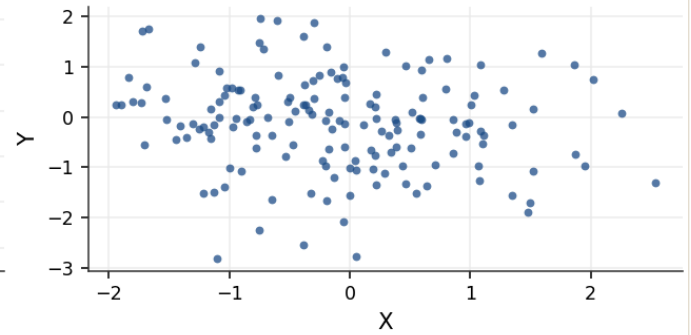
| What you see                                                          | What it means                                                                                     | What you do                                                                                                                                   |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Scatter is a straight cloud                                           | Linearity is met.                                                                                 | Pearson can be used.                                                                                                                          |
| Scatter is a U, or a clear curve                                      | A straight line is the wrong model.                                                               | Say linearity failed. Spearman does not repair a U.                                                                                           |
| Scatter is one ellipse                                                | Bivariate normality is met.                                                                       | Stay with Pearson if the other pictures also pass.                                                                                            |
| Scatter is two clumps, or not an ellipse                              | Bivariate normality is not met.                                                                   | If the pattern is still a steady rise or fall, tick Spearman or Kendall and report that coefficient.                                          |
| Residual plot: vertical spread stays similar across fitted values     | Homoscedasticity is met. Fitted means the predicted Y.                                            | Stay with Pearson.                                                                                                                            |
| Residual plot opens into a fan                                        | Homoscedasticity is not met.                                                                      | Report Spearman or Kendall, and say the residual plot was a fan.                                                                              |
| Residual spread is wider in the middle, with few points at the ends   | The plot is ambiguous.                                                                            | Write that it is ambiguous, which side you choose, and why.                                                                                   |
| Residuals trend up or down                                            | The straight line is systematically off. That is failed linearity.                                | Do not call this a fan.                                                                                                                       |
| Pearson $r$ , $p < .05$                                               | Significant relationship. The sign is the direction. $ r $ is the size.                           | Write both the significance and the size.                                                                                                     |
| Pearson $r$ , $p > .05$                                               | Not significant. $ r $ can still be medium. $r = .35$ is medium. $p = .051$ is not significant.   | Write both sentences.                                                                                                                         |
| The rows are ordered in time, such as sleep today with sleep tomorrow | jamovi will still print $r$ . The independence assumption is about the design, and it has failed. | Say the correlation is not appropriate because the rows are time-linked. The names for that problem are autocorrelation, AR, MA, ARMA, ARIMA. |

## Scatterplot: how big is $r$ ? Left = positive. Right = negative.

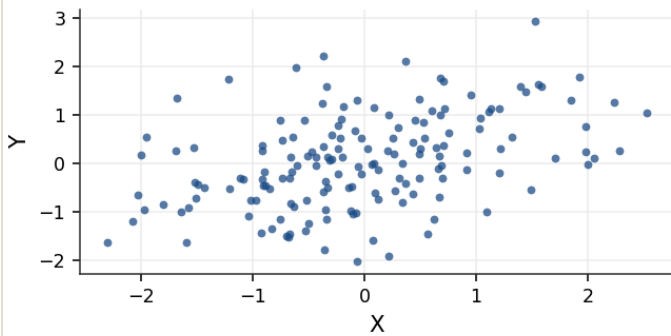
**Positive**  
 **$r = 0$**   
**No relationship. Round cloud.**



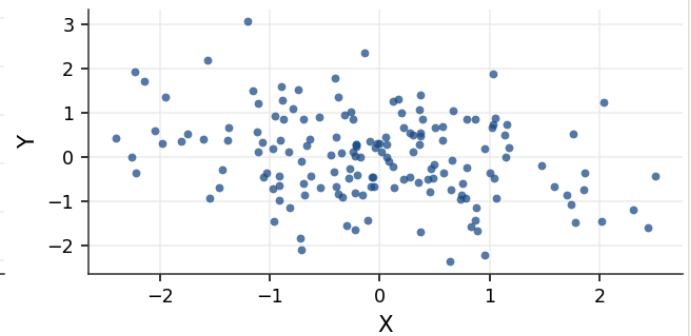
**Negative**  
 **$r = 0$**   
**No relationship. Round cloud.**



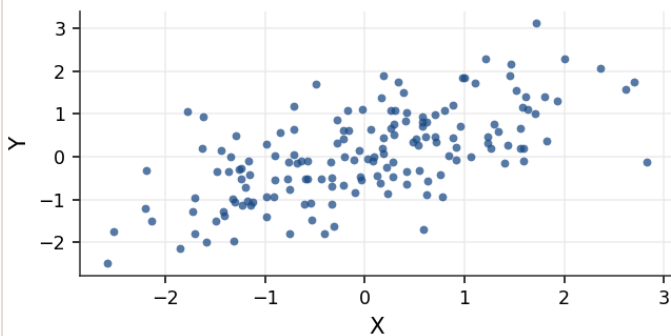
**Positive**  
 **$r = .33$  or  $-.33$**   
**Medium (Cohen  $\geq .30$ ). Mild tilt.**



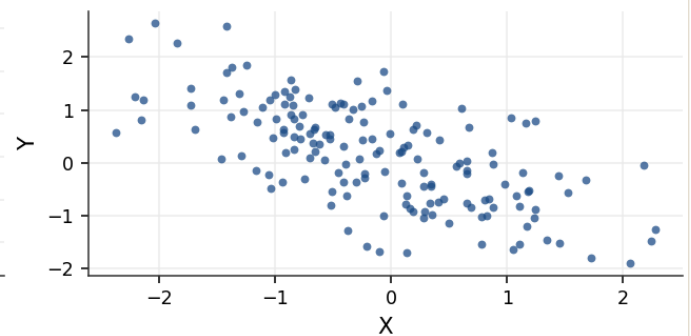
**Negative**  
 **$r = .33$  or  $-.33$**   
**Medium (Cohen  $\geq .30$ ). Mild tilt.**



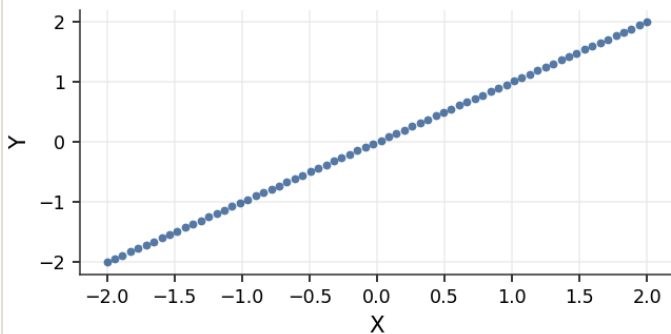
**Positive**  
 **$r = .66$  or  $-.66$**   
**Large (Cohen  $\geq .50$ ). Clear line, still scatter.**



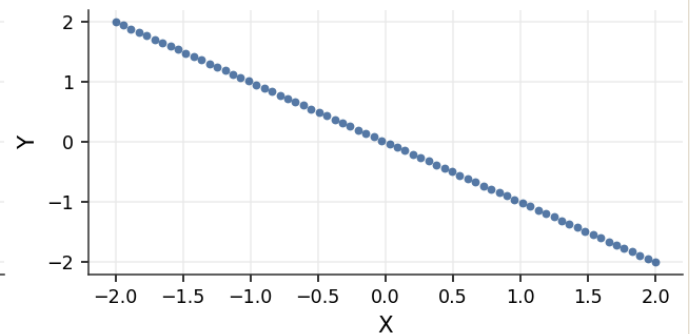
**Negative**  
 **$r = .66$  or  $-.66$**   
**Large (Cohen  $\geq .50$ ). Clear line, still scatter.**



**Positive**  
 **$r = 1$  or  $-1$**   
**Perfect. Every point sits on the line.**



**Negative**  
 **$r = 1$  or  $-1$**   
**Perfect. Every point sits on the line.**



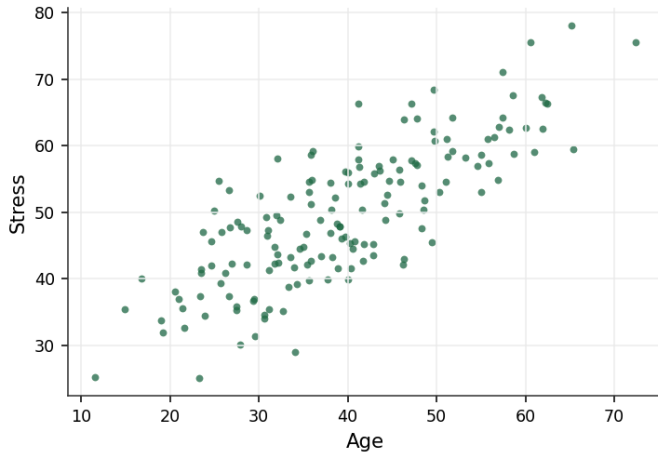
How big  $r$  looks. Left column is positive. Right column is negative. A round cloud is about 0. A mild tilt is about .33, medium. A clear line with scatter is about .66, large. A perfect line is 1 or  $-1$ .

## Scatterplot: the three assumptions you judge by eye

### Linearity MET

The cloud follows a straight line.

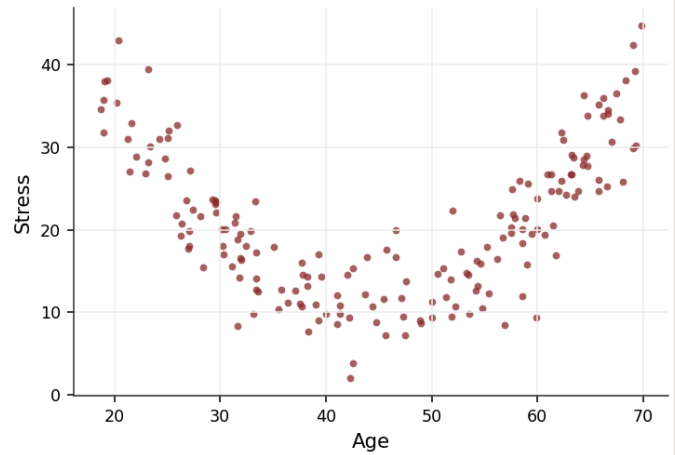
Pearson can be used if the other checks pass.



### Linearity VIOLATED

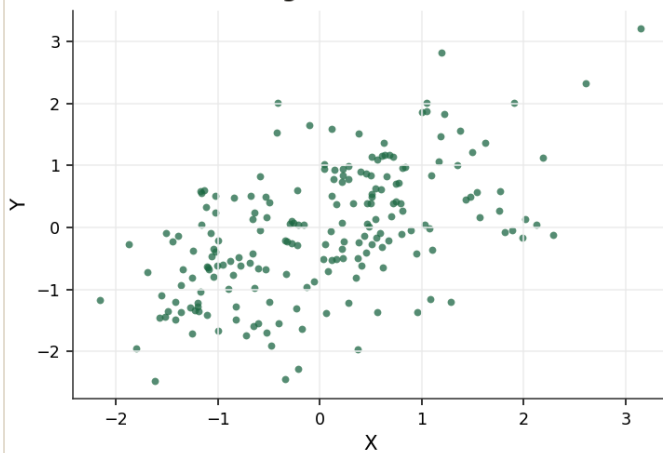
U-shape. Pearson is the wrong model.

Spearman also misses a U.



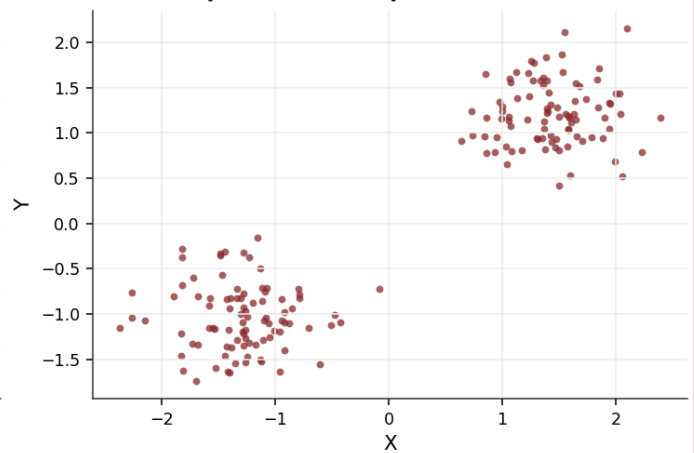
### Bivariate normality MET

One ellipse. Dense in the middle, thinner at the edges.



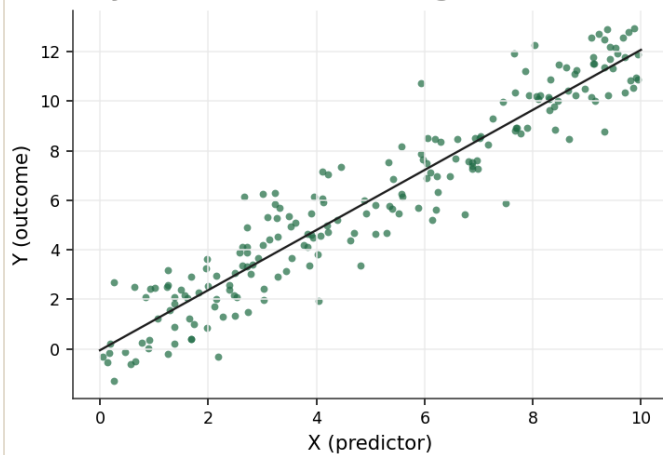
### Bivariate normality VIOLATED

Two clumps, not one ellipse.



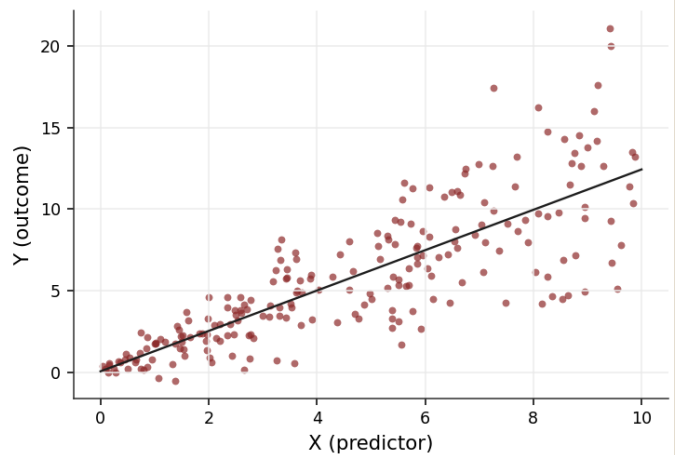
### Homoscedasticity MET

Vertical distance from the line stays about the same all along X.



### Homoscedasticity VIOLATED (fan)

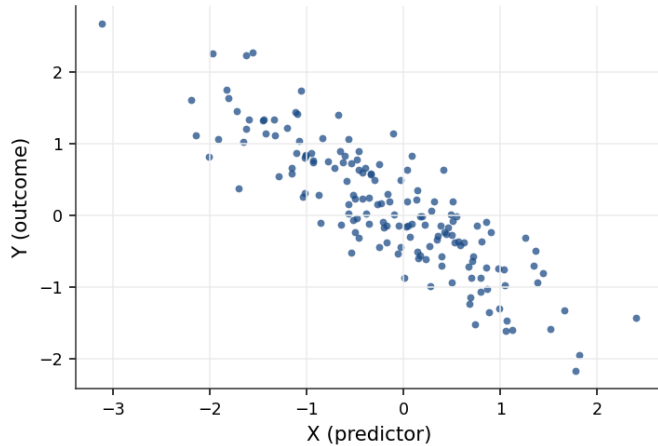
Spread grows as X increases. Do not use Pearson.



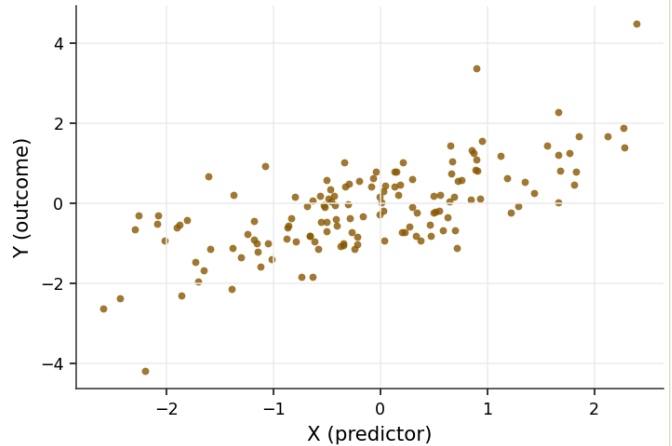
Three assumptions on the scatterplot. Linearity met = a straight cloud. Linearity violated = a U. Spearman does not repair a U. Bivariate normality met = one ellipse, dense in the middle. Violated = two clumps. Homoscedasticity met = the vertical distance from the line stays about the same all along X. Violated = a fan, the spread grows as X increases. A fan means do not use Pearson.

## Scatterplot: cases that are easy to misread

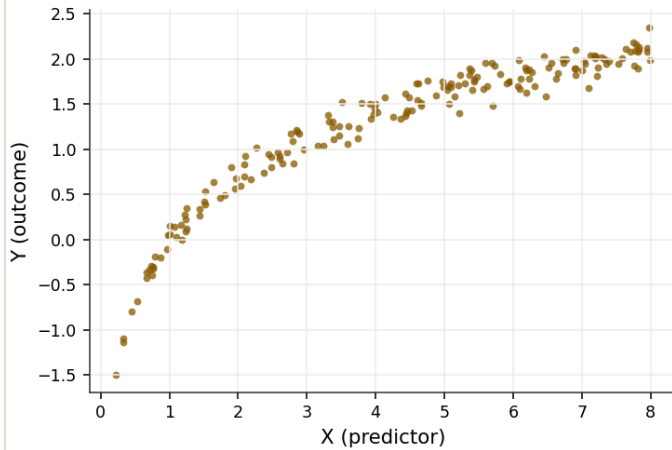
**Large NEGATIVE, assumptions OK**  
**r around  $-.85$ . Straight ellipse,**  
**even spread. Say negative and large.**



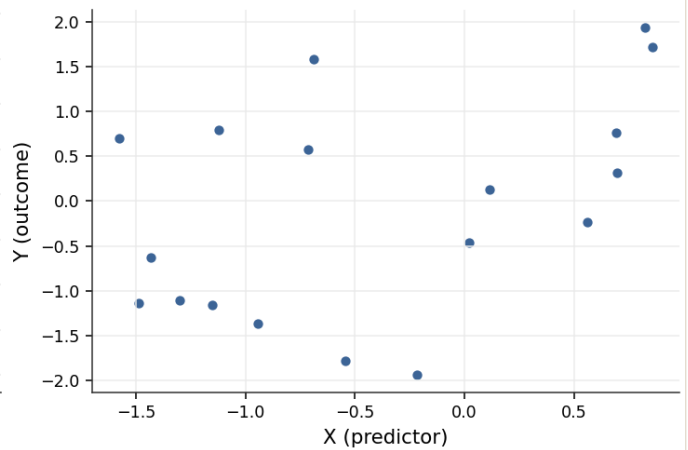
**A few outliers, cloud still linear**  
**Mention them. Two points do not**  
**fail the whole assumption.**



**Curved but always rising**  
**Pearson linearity is shaky.**  
**Spearman can describe this rise.**  
**This is not a U-shape.**



**Small n — do not over-call**  
**Only 18 points. If it is unclear,**  
**say it is unclear.**

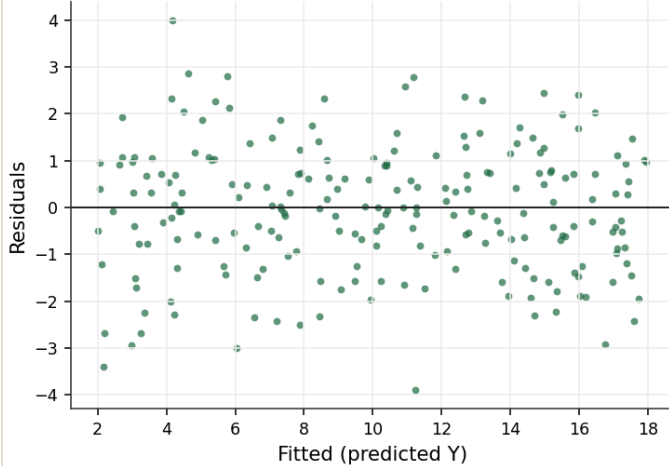


Easy to misread. A steep downward ellipse can still be linear, one ellipse, and even spread: say negative and large. A few outliers do not fail the whole assumption. A curve that always rises is not a U: Pearson linearity is shaky, and Spearman can describe that rise. With very few points, if the shape is unclear, say it is unclear.

## Residual plot: Fitted vs residuals

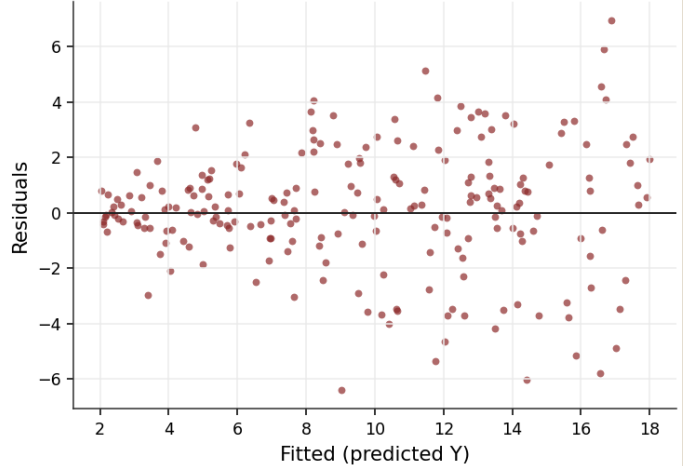
### MET

Vertical spread is similar across fitted values.



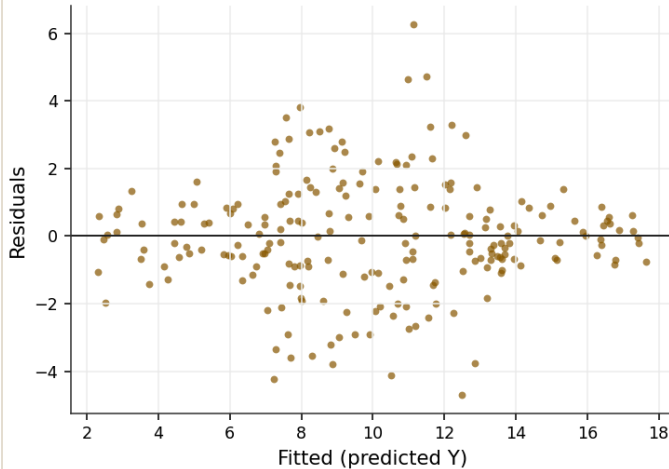
### VIOLATED — fan

Spread grows as fitted values grow.



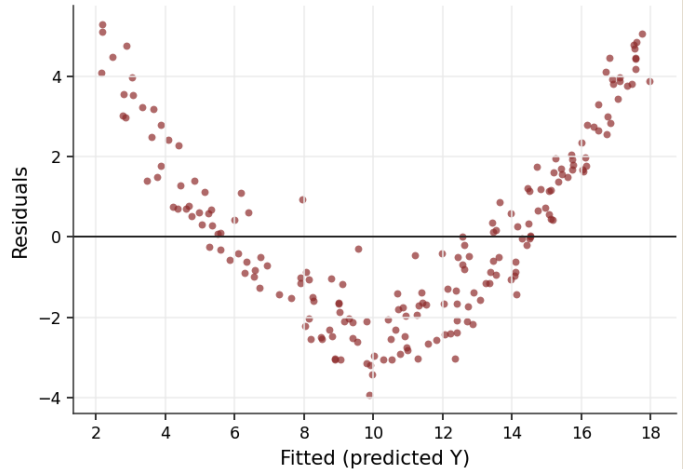
### AMBIGUOUS — wider in the middle

Ends have fewer points. Say so, then pick a side.



### Bias, not a fan

Residuals curve. The straight line is wrong. This is failed linearity.



Residual plot, fitted versus residuals. This is the homoscedasticity picture from Linear Regression. Met = vertical spread stays similar. Fan = spread grows as fitted values grow, so homoscedasticity is not met. Wider in the middle = ambiguous: say so, pick a side, give the reason. A curve = failed linearity, not a fan.

There was a significant / not significant [positive / negative] [small / medium / large] relationship between [X] and [Y],  $r = \_\_\_$ ,  $p = \_\_\_$ . As [X] increased, [Y] increased / decreased. The scatter was a straight cloud / a curve. The cloud was one ellipse / not one ellipse. The residual plot had similar vertical spread / a fan / an ambiguous spread. [If you changed coefficient:] Spearman's rho / Kendall's tau-b =  $\_\_\_$ ,  $p = \_\_\_$ , because [the Q-Q or the residual plot] failed.

# Partial Correlation

**Click the simple correlation first**, so you have the  $r$  before any control. Then Analyses → Regression → Partial Correlation. Variables = the two columns in the question. Control Variables = the column you hold constant. Tick Pearson. Under Correlation type, jamovi shows two choices.

**Partial** removes the control column from both variables. The two off-diagonal cells match.

**Semipartial** removes the control column from only one variable. The two off-diagonal cells can be different numbers. Copy the cell that removes the control from the predictor, and name that variable.

| What happens to $r$ after the control                       | The name of that pattern |
|-------------------------------------------------------------|--------------------------|
| The controlled $r$ is weaker and its $p$ is still below .05 | Partial confounder       |
| The controlled $r$ 's $p$ is above .05                      | Confounder               |
| The controlled $ r $ is stronger, same sign                 | Suppressor               |
| The sign flips                                              | Negative suppressor      |

The simple relationship between  $[X]$  and  $[Y]$  was [positive / negative] and [size],  $r = \_\_\_$ ,  $p = \_\_\_$ . After controlling for  $[Z]$ , the partial / semi-partial correlation was  $r = \_\_\_$ ,  $p = \_\_\_$ . The relationship became weaker / was no longer significant / became stronger / flipped sign. This pattern is a partial confounder / confounder / suppressor / negative suppressor.

Pattern from class: simple  $r = .664$ ,  $p < .001$ , positive large. After the control, partial  $r = -.618$ ,  $p < .001$ . The sign flipped and the new  $p$  is still below .05. That is a negative suppressor.

## One-Way ANOVA — different people, 3 or more groups

**Two levels in the grouping column.** Use the independent t screen. One-Way ANOVA is for 3 or more levels.

**The Post-Hoc Test button named None prints no pair table.** jamovi's own setting says None shows no post-hoc tests. It is not a results table.

**There is no  $\eta^2$  tick box on One-Way ANOVA.** The  $\eta^2$ , partial  $\eta^2$ , and  $\omega^2$  boxes are on Analyses → ANOVA → ANOVA.

**There is no button named Fisher's LSD.** The uncorrected pair table is on ANOVA → ANOVA → Post Hoc Tests → Correction → **No correction**.

**Click, screen 1.** Analyses → ANOVA → One-Way ANOVA. Dependent Variables = the score. Grouping Variable = the group column. Tick both **Don't assume equal (Welch's)** and **Assume equal (Fisher's)**. Tick Descriptives table. Assumption Checks: Homogeneity test, Q-Q Plot. Under Post-Hoc Tests, tick Test results (t and df) and Report significance.

**Click, screen 2, for  $\eta^2$  and for No correction.** Analyses → ANOVA → ANOVA. Dependent Variable = the same score. Fixed Factors = the same group column, and no second factor. Effect Size: tick  $\eta^2$ . Assumption Checks can stay off if you already have Levene and the Q-Q plot on screen 1. Post Hoc Tests: move the group column into the box. Under Correction, the choice you tick depends on the case below.

### Which F row

| Levene p on screen 1 | What it means      | Copy this F                                                               |
|----------------------|--------------------|---------------------------------------------------------------------------|
| Above .05            | Equal variances.   | The Fisher's row. df1 = number of groups – 1. df2 = N – number of groups. |
| Below .05            | Unequal variances. | The Welch's row. Its df2 is not N – number of groups.                     |

### What that F means

| Fisher's or Welch's p, from the row you chose | What it means                                                                     | Post hoc                                                                   |
|-----------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| $p > .05$                                     | F is not significant. The test did not find a difference among the means.         | Stop. Do not describe a winning pair. You can leave Post-Hoc Test on None. |
| $p < .05$                                     | F is significant. The means are not all equal. The F row does not say which pair. | Use the post hoc case that matches Levene and the number of groups.        |

### Which post hoc, after a significant F

| Case                                              | What you click                                                                                                | What you copy                                                                                                                       |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Levene $p > .05$ , and 3 groups                   | Screen 2. Correction: <b>No correction</b> . Untick Tukey if it is already on, so the table has one p column. | Each pair's t, df, and p from the No correction column. This is the uncorrected follow-up. With 3 groups there are 3 pairs.         |
| Levene $p > .05$ , and 4 or more groups           | Screen 1. Post-Hoc Test: <b>Tukey (equal variances)</b> .                                                     | Each pair's t, df, and p from the Tukey table. Four groups make 6 pairs, so the uncorrected p-values are the wrong column.          |
| Levene $p < .05$ , any number of groups from 3 up | Screen 1. Post-Hoc Test: <b>Games-Howell (unequal variances)</b> . The F you already copied is Welch's.       | Each pair from the Games-Howell table. Games-Howell is on the One-Way ANOVA screen. It is not in the ANOVA → ANOVA correction list. |
| A pair's $p < .05$                                |                                                                                                               | Those two groups differ. The higher mean is the direction.                                                                          |
| A pair's $p > .05$                                |                                                                                                               | Those two groups do not differ significantly. Say that. Do not say they are equal.                                                  |
| F is significant and every pair's p is above .05  |                                                                                                               | Write both facts. Do not invent a winning group.                                                                                    |

**No correction** means jamovi prints the pairwise p-values with no adjustment for the number of pairs. The slides call this Fisher's LSD. The words on the button are No correction.

**Tukey** means the p-values are adjusted for the set of pairs, for the equal-variance case. A Tukey p is not the same number as the No correction p for that pair. Copy the column that matches the case.

**Games-Howell** means the pair test that does not assume equal variances. Use it with the Welch's F row.

**$\eta^2$  on screen 2** is SS for the grouping factor divided by total SS. With only one factor in Fixed Factors, partial  $\eta^2$  prints the same number as  $\eta^2$ . Tick  $\eta^2$  and write  $\eta^2$ . There is no size label. Write the proportion.

Means (SDs): [group A] \_\_ (\_\_), [group B] \_\_ (\_\_), [group C] \_\_ (\_\_). The Q-Q plot followed the line / bent, so normality was met / not met. Levene's test was  $F(\\_, \\_) = \\_, p = \\_$ , so variances are equal / unequal. The [Fisher's / Welch's] one-way ANOVA was significant / not significant,  $F(\\_, \\_) = \\_, p = \\_, \eta^2 = \\_$ . [If F is significant:] Post hoc [No correction / Tukey / Games-Howell]: [pair]  $t(\\_) = \\_, p = \\_$ . [Say which pairs differ and which p is above .05.]

Pattern from class: Levene  $p = .266$ , above .05, three groups, Fisher's  $p < .001$ , so F is significant and the follow-up is No correction.  $\eta^2 = 0.71$ . One pair, anxifree versus placebo, has  $p = .150$ , which is above .05, so that pair is not significant.

## Repeated Measures ANOVA — same people, 3 or more columns

**Two columns from the same people.** Use Paired Samples T-Test. Repeated Measures ANOVA is for 3 or more columns.

**Click.** Analyses → ANOVA → Repeated Measures ANOVA. Repeated Measures Factors: type a name, set the number of levels, label each level. Drag each score column into its level. Leave Between Subject Factors and Covariates empty.

Effect Size: tick **Partial  $\eta^2$** .  $\eta^2$  and Generalised  $\eta^2$  are different checkboxes. For this one-way repeated design, report the column headed Partial  $\eta^2$ .

Assumption Checks: tick **Sphericity tests** and **Q-Q Plot**. Under Sphericity corrections, None is the box that is already available. Tick Greenhouse-Geisser or Huynh-Feldt only in the cases below, so the matching F row appears.

Post Hoc Tests: move the repeated factor into the box. jamovi starts with **Tukey** ticked. Untick Tukey and tick **Holm** unless the question names another correction.

| What you see                                                   | What it means                                                           | Which F row you copy                                                                                             |
|----------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Q-Q follows the line                                           | Normality is met.                                                       | Write that.                                                                                                      |
| Q-Q bends                                                      | Normality is not met.                                                   | Write that, then continue with Mauchly.                                                                          |
| Mauchly $p > .05$                                              | Sphericity is met. The sphericity test did not find a violation.        | The None row under Sphericity corrections. Do not apply $\epsilon$ .                                             |
| Mauchly $p < .05$ , and Greenhouse-Geisser $\epsilon > .75$    | Sphericity is not met. $\epsilon$ is the number in the Mauchly table.   | Tick Huynh-Feldt. Copy the Huynh-Feldt F, df, and p.                                                             |
| Mauchly $p < .05$ , and Greenhouse-Geisser $\epsilon \leq .75$ | Sphericity is not met, and the departure is larger.                     | Tick Greenhouse-Geisser. Copy that F, df, and p. Greenhouse-Geisser is the stronger correction of the two boxes. |
| That F has $p < .05$                                           | F is significant. The condition means are not all equal.                | Then read the Holm pair table.                                                                                   |
| That F has $p > .05$                                           | F is not significant.                                                   | Stop. Do not name a winning condition.                                                                           |
| Holm $p < .05$ for a pair                                      | Those two conditions differ.                                            | Name the higher mean.                                                                                            |
| Holm $p > .05$ for a pair                                      | Those two conditions do not differ significantly.                       | Say so, including a p of .058.                                                                                   |
| F is significant and every Holm p is above .05                 | The overall test found a difference somewhere, and no pair reached .05. | Write both facts.                                                                                                |

**Partial  $\eta^2$**  is the effect-size column that leaves out the between-person variance. That is the column to write for this screen. jamovi does not print a small / medium / large label next to it.

**No correction, Tukey, Scheffe, Bonferroni, Holm** are five p-value adjustments for the same pairs. They are not the same p. Holm is the one to tick for this test unless the question names another. Bonferroni and Scheffe are stricter. No correction is the raw pair p.

For 3 conditions, Mauchly's df is usually 2. If jamovi prints a different df, copy the df on the screen.

[Condition A] M = \_\_ (SD = \_\_), [B] M = \_\_ (SD = \_\_), [C] M = \_\_ (SD = \_\_). The Q-Q plot followed the line / bent, so normality was met / not met. Mauchly's test was  $\chi^2(\_) = \_, p = \_,$  so sphericity was met / not met. [If not met: Greenhouse-Geisser  $\epsilon = \_.$  Because  $\epsilon$  is above / not above .75, the Huynh-Feldt / Greenhouse-Geisser row was used.] The repeated-measures ANOVA was significant / not significant,  $F(\_, \_) = \_, p = \_,$  partial  $\eta^2 = \_.$  Post hoc Holm: [pair]  $t(\_) = \_, p = \_.$  [If every pair is above .05, say the omnibus test was significant and no pair reached .05.]

Pattern from class: Mauchly  $p = .720,$  above .05, so the None correction row.  $F p = .013,$  below .05, so F is significant. partial  $\eta^2 = .581.$  The pair p-values .381, .058, and .102 are all above .05, so no pair is significant.

## One page of cases

| You see                                          | You report                                                    |
|--------------------------------------------------|---------------------------------------------------------------|
| 2 groups, different people, Levene $p > .05$     | Student's t, d, CI, who is higher                             |
| 2 groups, different people, Levene $p < .05$     | Welch's t, d, CI, who is higher                               |
| 2 columns, same people                           | Paired t. No Levene sentence.                                 |
| Two scores, pictures pass, $p < .05$             | Pearson r, sign, size                                         |
| Two scores, $p > .05$                            | Not significant, and still the size of  r                     |
| Fan, or a non-ellipse that still rises or falls  | Spearman or Kendall, and the reason                           |
| U-shape                                          | Linearity failed. Do not switch to Spearman.                  |
| A third score is held constant                   | Simple r, then Partial, then the pattern name                 |
| The control is removed from one column only      | Semipartial, and name which column                            |
| 3+ groups, Levene $p > .05$ , Fisher's $p > .05$ | Fisher's F and $\eta^2$ . No winning pair.                    |
| 3 groups, Levene $p > .05$ , Fisher's $p < .05$  | Fisher's F, $\eta^2$ , and the No correction pairs            |
| 4+ groups, Levene $p > .05$ , Fisher's $p < .05$ | Fisher's F, $\eta^2$ , and Tukey pairs                        |
| 3+ groups, Levene $p < .05$ , Welch's $p < .05$  | Welch's F, $\eta^2$ from the ANOVA screen, Games-Howell pairs |
| 3+ columns, same people, Mauchly $p > .05$       | Uncorrected F, partial $\eta^2$ , Holm pairs                  |
| Same, Mauchly $p < .05$ , $\epsilon > .75$       | Huynh-Feldt F, partial $\eta^2$ , Holm pairs                  |
| Same, Mauchly $p < .05$ , $\epsilon \leq .75$    | Greenhouse-Geisser F, partial $\eta^2$ , Holm pairs           |
| F $p < .05$ and every pair $p > .05$             | Say both. Do not pick a winner.                               |